THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

National Mathematics Talent Contests – Screening Test

Mathematics Olympiad at **Junior Level** *for IX & X Classes*

Bhaskara Contest – Saturday, August 30, 2003

Note: Read carefully the instructions given in the Response Sheet.
Duration of the test 2pm to 4p.m

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1. A fully loaded truck weighs t tons. The weight of the empty truck is w tons. The weight, when the truck is loaded to half its capacity, would be

   A) $\frac{t}{2}$ B) $\frac{t}{2}+w$ C) $\frac{t-w}{2}$ D) $\frac{t+w}{2}$

2. When the length of side of a cube is increased by 10% the volume of the cube increases by

   A) 10 % B) 22.1% C) 33.1% D) 30%

3. 27 metal balls each of radius r are melted together to form one big sphere of radius R. What is the ratio the surface area of the big sphere is to that of a ball?

   A) 9 B) 3 C) $\sqrt{27}$ D) $\sqrt{3}$

4. The 100^{th} root of $10^{(10^{10})}$ is

   A) $10^{8^{10}}$ B) $10^{10^{8}}$ C) $\left(\sqrt{10}\right)^{(\sqrt{10}\,)^{10}}$ D) $10^{(\sqrt{(10)})^{\sqrt{10}}}$

5. A regular polygon has 27 diagonals. The number of sides is

   A) 10 B) 9 C) 11 D) 12

6. The average age of an eleven-member cricket team reduces by 2 when two players of ages 30 and 32 leave the team and two new players of equal age join it. The age of each of the newcomers is

   A) 20 B) 62 C) 12 D) 22

7. The product of the ages of three sisters is 36. The sum of their ages is a prime number. The youngest sister likes ice cream. The product of the ages of the

8. C is the set of all sums of cubes of three consecutive natural numbers. Then every element of C is divisible by

A) 2 B) 3 C) 4 D) 5

9. Three people each think of a number which is the product of two different primes. Which of the following could be the product of three numbers which each one of them thought of?

A) 120 B) 144 C) 12100 D) 3000

10. A number n leaves the same remainder while dividing 5814, 5430, 5958. What is the largest possible value of n?

A) 96 B) 12 C) 48 D) 192

11. One hundred and twenty Students take an examination which is marked out of a total of 100 (with no fractional marks). No three students are awarded the same mark. What is the smallest possible number of pairs of students who are awarded the same mark?

A) 9 B) 10 C) 20 D) 19

12. 36 coins are divided among A, B, C and D such that B takes more than A, C more than B and D more than C. If the difference of the number of coins between A and B is 1, between B and C is 9 and between C and D is 7, then the number of coins A got is

A) 12 B) 8 C) 6 D) 2

13. The value of k for which $(x+2)$ is a factor of $(x+1)^7+(3x+k)^3$ is

A) -7 B) 7 C) -1 D) $-6-3^{(7/3)}$

14. Given that $x_1, x_2, x_3, \ldots, x_{15}, x_{16}$ are positive real numbers such that

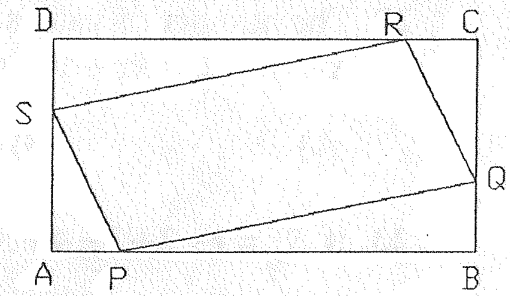
$$\frac{x_1}{x_2}=\frac{x_2}{x_3}=\frac{x_3}{x_4}\cdots=\frac{x_{15}}{x_{16}},$$

if $x_1+x_2+x_3+x_4=20$, $x_5+x_6+x_7+x_8=320$, then $x_{13}+x_{14}+x_{15}+x_{16}$ is

A) 20480 B) 19680 C) 40960 D) 81920

15. If $x=9+4\sqrt{5}$ and $xy=1$ then $\frac{1}{x^2}+\frac{1}{y^2}$ is

16. Points P, Q, R, and S divide the sides of a rectangle $ABCD$ in the ratio 1:2 as shown in the figure. What fraction of the area of the rectangle is the area of the parallelogram $PQRS$?

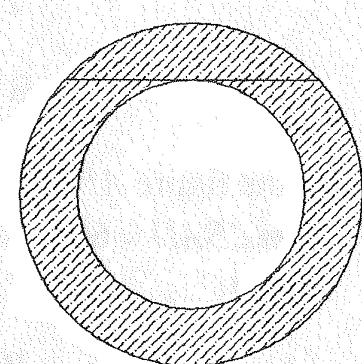


A) $\frac{2}{5}$　　B) $\frac{3}{5}$　　C) $\frac{4}{9}$　　D) $\frac{5}{9}$

17. What is the last digit in the finite decimal representation of the number $\frac{1}{5^{2003}}$?

A) 2　　B) 4　　C) 6　　D) 8

18. The diagram shows two concentric circles. The chord of the large circle is a tangent to the small circle and has length $2p$.
What is the area of the shaded region?

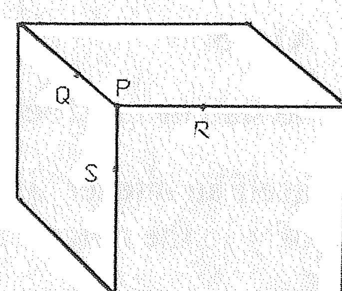


A) πp^2　　B) $2\pi p^2$　　C) $3\pi p^2$　　D) $4\pi p^2$

19. When rounded to 3 significant figures the number x is written as 1000. What is the largest range of possible values of x?

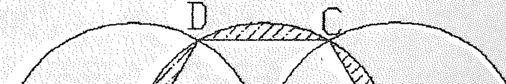
A) $999 \leq x < 1001$　　B) $995 \leq x < 1005$
C) $990 \leq x < 1010$　　D) $999.5 \leq x < 1005$

20. P is a vertex of a cuboid and Q, R and S are three points on the edges as shown.
$PQ = PR = 2\ cm$ and $PS = 1\ cm$.
What is the area of ΔQRS (in cm^2)?



A) $\frac{\sqrt{15}}{4}$　　B) $\frac{5}{2}$　　C) $\sqrt{6}$　　D) $2\sqrt{2}$

21. AB is a line segment of length 4 cm.



22. The number of values of k for which the system of equations
$x+y=2, \quad kx+y=4, \quad x+ky=5$ has at least one solution is

A) 0 B) 1 C) 2 D) 3

23. If $S_n = 1-2+3-4+5-6+7-8+\ldots$ (up to n terms), then $S_{2002} - S_{2003} + S_{2004}$ is

A) 3005 B) -1001 C) -3005 D) 1001

24. The sum of all the five digit numbers that can be formed using the digits 1,2,3,4, and 5 (repetition of digits not allowed) is

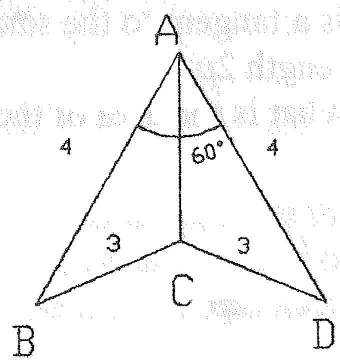
A) 360000 B) 660000 C) 366000 D) 3999960

25. If $y \neq 0$, then the number of pairs (x, y) such that

$$x + y + \frac{x}{y} = \frac{1}{2} \text{ and } (x+y)\frac{x}{y} = -\frac{1}{2} \text{ is}$$

A) 1 B) 2 C) 0 D) countless

26. In the figure $ABCD$ is a non-convex quadrangle.
If $m\angle BAD = 60^\circ$, $AB = 4 = AD$ and BC =3=DC
then AC is



A) $2\sqrt{3}-\sqrt{5}$ B) $2\sqrt{3}-5$
C) $2\sqrt{3}+\sqrt{5}$ D) $2\sqrt{3}+5$

27. In how many different ways can I circle letters in the grid shown so that there is exactly one circled letter in each row and exactly one circled letter in each column?

A	B	C	D	E
F	G	H	I	J
K	L	M	N	O
P	Q	R	S	T
U	V	W	X	Y

A) 15 B) 24 C) 60 D) 120

28. The statement "There are exactly four integer values of n for which $\frac{2n+y}{n-2}$ is itself an integer" is true for certain values of y only. For how many values of y in the range $1 \leq y \leq 20$ is the statement true?

A) 0 B) 7 C) 8 D) 10

29. If α, β are the roots of $\left(x-\sqrt{2003}\right)\left(x-\sqrt{2002}\right)=\sqrt{2004}$, then the roots of $(x-\alpha)(x-\beta)=\sqrt{2004}$ are

A) $\sqrt{2003}, \sqrt{2002}$ B) $\sqrt{2002}, \sqrt{2004}$